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Formation and evolution of the Masirah ophiolite constrained by paleomagnetic study of volcanic rocks

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Abstract

The extrusive rocks of the lower ophiolitic nappe of Masirah Island (Oman) were paleomagnetically studied. Four mean directions of magnetization were isolated, one at low temperature, the others at high temperature. The low-temperature component, found in most samples, corresponds to a recent viscous and/or chemical remagnetization. The high-temperature component, found at three sites in the Centre area, is interpreted as a possible primary magnetization acquired during extrusion or, more probably, as an early remagnetization related to subsequent hydrothermal alteration and indicates formation of the Masirah ophiolite around a paleolatitude of 40°S, close to the present position of the West Somali basin. A second high-temperature component, found in three other sample sites in the Hakl and in the Thumi areas, is probably a chemical remagnetization related to the emplacement of the upper ophiolite nappe in late Maastrichtian to Paleocene times. Finally a last high-temperature component, found only in the two sites from Naft area, could be a post-tectonic remagnetization acquired in Oligocene or Miocene times or represents differential tilting of pillows and sedimentary blocks. The high-temperature components are usually fairly scattered due to the very complex tectonic history of the area. Nevertheless, the relation between the defined paleopoles and the Indian polar wander is unambiguous and the evolution of the Masirah ophiolite from formation up to the emplacement of the second ophiolitic nappe is clearly related to the northward movement of the Indian plate. This new paleomagnetic study rules out the possibility of a common or related origin for the Masirah and the Semail ophiolites.

1. Introduction

Masirah is a 650-km^2-large island located close to the Oman coast at the margin of the Indian Ocean (Fig. 1). Masirah Island consists nearly exclusively of ancient oceanic floor material and forms the largest outcrop along the "Eastern Ophiolite Belt" of Oman, to which belong also some fragments of oceanic crust and associated sediments of similar type at Ras Madrakah, Ras Jibsch and in the Batain region (see Shackleton et al., 1990). The earliest geological investigations on Masirah were undertaken by Carter (1847) for economic studies of copper mineralization. Lees (1928) compared the igneous rocks of

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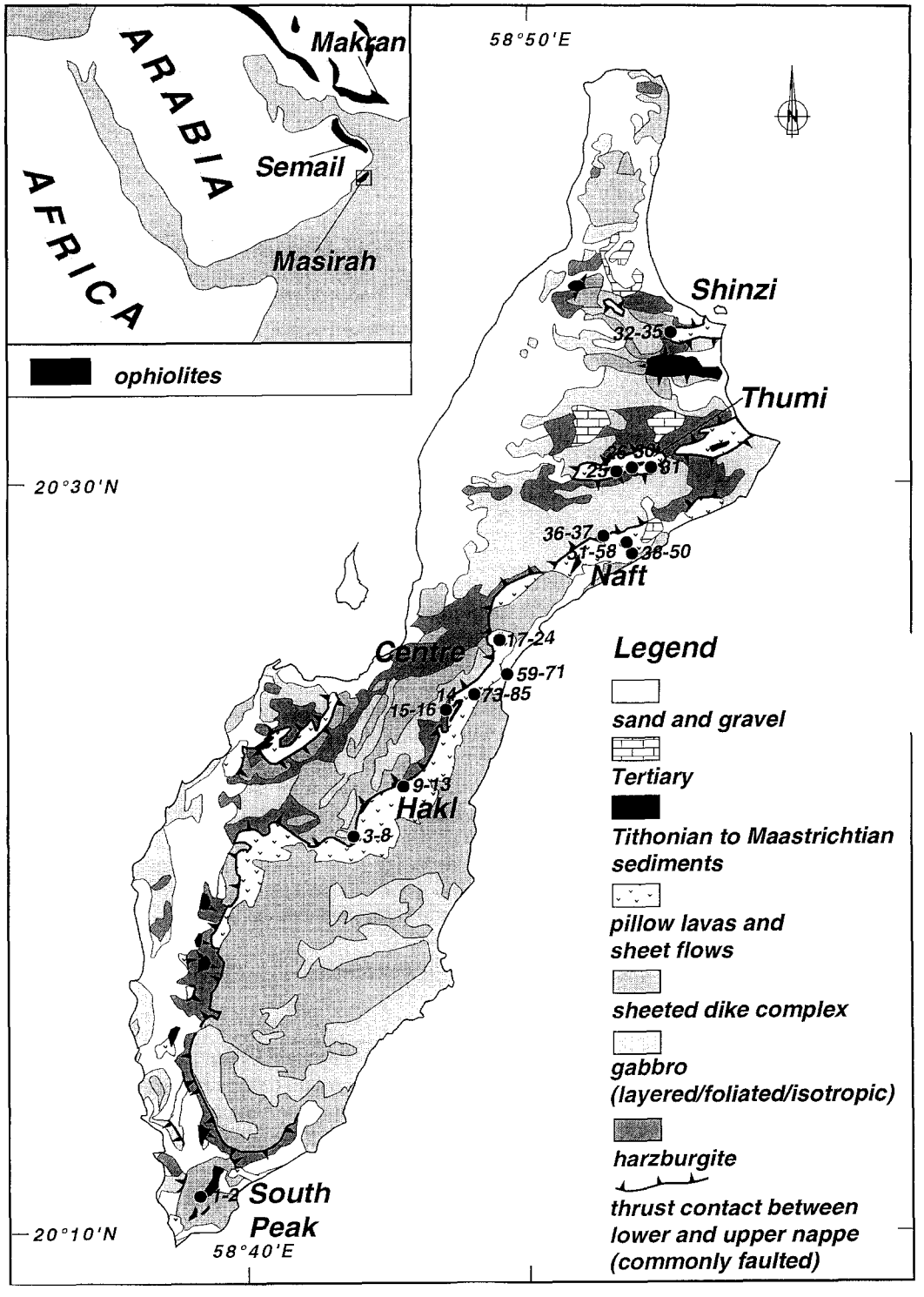


Fig. 1. Geological map of Masirah Island showing the two ophiolite nappes and localities of the pillow lavas sampled at the top of the lower nappe. The inset is an overview of the Gulf area with location of Masirah Island and other Tethyan ophiolites.

Masirah with the Semail ophiolite and dated the Tertiary cover as Middle Eocene. Moseley (1969) produced a geological map of the island and, in a study on the Semail ophiolite, Glennie et al. (1974) included a geological map at 1:500,000 scale of the Masirah island, where they noted that the relationship between the sediments found on Masirah island and the Hawasina sediments associated with the Semail Ophiolite was not understood. Abbotts (1978) studied the granites associated with the ophiolite and the metamorphism and chemical composition of the extrusive sequence (Abbotts, 1979, 1981). Beurrier (1987) reported Tithonian to Hauterivian ages of radiolarian cherts overlying lavas, and Smewing et al. (1991) radiometrically dated biotite from granites and hornblende from gabbroic rocks with K-Ar methods. Both biotites and hornblendes showed a widely overlapping spread between 124 ± 4 and 158 ± 9 Ma, interpreted as obduction and formation ages, respectively. The age results show that the Masirah ophiolites are older than the Upper Cretaceous Semail Ophiolite (e.g., Allemann and Peters, 1972; Glennie et al., 1974; Tilton et al., 1981; Tippit et al., 1981), and Moseley and Abbotts (1979) proposed an origin in the Indian Ocean. Le Métour et al. (1992) produced a 1:250,000 geological map of Masirah and described the folded Upper Cretaceous Fayah Formation from the island.

Paleomagnetic studies on ophiolites, predominantly on the extrusive members and associated sediments, are widely used for paleogeographic relocation of ancient ocean floor fragments (e.g., Thomas et al., 1988; Perrin et al., 1994; Mubroto et al., 1994). The magnetic mineralogy of lava may undergo large alterations during seafloor and emplacement-related metamorphism. However, since seafloor alteration takes place shortly after the extrusion of the lava, the first secondary ore mineral formation of magnetite or maghemite after Ti-magnetite will record the same field period (e.g., Prévot et al., 1979). Although the magnetic intensity is generally strongly reduced, the magnetization direction remains. An off-axis metamorphism in greenschist facies may completely reset the magnetic signal to the prevailing magnetic field orientation.

In the present study on lava from the lower Masirah ophiolitic nappe, we intend to support evidence for a complex structural evolution by paleomagnetic methods and to reconstruct the origin of the lower Masirah nappe. This study is a contribution to the Masirah project led by Prof. Peters, University of Bern.

2. Regional setting

Recent detailed mapping at a scale of 1:50,000 has revealed that Masirah is composed of two ophiolitic nappes (Fig. 1; Marquer et al., 1995). A classic ophiolite sequence (Penrose, 1972) can easily be recognized, although the two nappes are strongly affected by pre- and post-nappe faulting. The two ophiolite nappes are composed of the following lithologic units: the base of the lower ophiolite nappe is nowhere exposed and the series starts with layered and foliated gabbros containing abundant melanotroctolitic to wehrlitic intrusions which bend and disturb the gabbro foliation. Small harzburgite outcrops are very locally exposed. The sheeted dike complex consists of half a meter (to few meters) thick dolerite dikes with no or weakly developed chilled margins. The dike swarms cut, in most localities, the layered gabbros. The sheeted dike complex grades upwards into a sheet flow and pillow lava sequence which itself is overlain by pre-obduction Tithonian to Maastrichtian deep-sea to shallow-water sediments with alkaline volcanic or volcanoclastic interlayers (Immenhauser, in prep.). The oldest sediments of Tithonian age give the age of the tholeiitic extrusives. The youngest flysch-type sediments of the Fayah Formation (Le Métour et al., 1992) are of late Maastrichtian age (Immenhauser, in prep.) which gives a maximum age for the thrusting of the upper nappe onto the lower (Marquer et al., 1995). The Fayah Formation contains boulders derived from continental crystalline basement. The sediments at the top of the lower nappe are folded and the underlying pillow lavas are imbricated. As a result, the pillow lavas and sediments may be steeply dipping.

The upper nappe was detached around the Moho (first occurrence of gabbros) and starts with harzburgites and few dunites. The sequence grades upwards into layered or foliated gabbros containing deformed wehrlitic to melanotroctolitic intrusions. The thickness of the crustal sequence is highly variable and the sheeted dike complex commonly intrudes mantle

peridotites. Pillow lavas are only found very locally in the upper nappe and represent the highest exposed levels. Granitoid intrusions, associated with isotropic and pegmatoid gabbros are found in peridotites and gabbros. The upper nappe is, in the northern part of the island, locally covered by Paleocene to Eocene shallow-water carbonates (Le Métour et al., 1992; Briner and Stucky, 1994), separated by a lateritic horizon from the underlying ophiolite.

The sheeted dike complex is dominantly ENE trending (Moseley and Abbotts, 1979). The so-called "melange" of the Masirah fault (Moseley and Abbotts, 1979; Shackleton and Ries, 1990) is the result of interference between the subhorizontal thrust plane, multiply offset by normal faults, and intersection with a flat topography (Marquer et al., 1995). A composite profile of the two ophiolite nappes with the sandwiched sediments below the thrust contact is schematically shown in Fig. 2.

Important stages in the geological evolution of the lower Masirah nappe are:

(1) formation of tholeiitic extrusives (lower nappe) in the Tithonian;

(2) alkaline volcanism associated with subvolcanic trachytes around 123 ± 2 Ma (Nägler and Frei, 1994; Immenhauser, in prep.). The volcanic rocks are locally deposited on serpentinites or gabbros and covered by Cretaceous shallow-water carbonates (Immenhauser, in prep.) which indicates important tectonic movements and uplift;

(3) southward directed overthrusting of the upper nappe between the Maastrichtian and the Eocene causing folding in the sedimentary cover of the lower nappe.

(4) post-Eocene (probably Miocene) NNE–SSW-oriented horst and graben formation related to extension and followed by N–S- to NNW–SSE-oriented normal faulting. These late events are responsible for the different levels of Tertiary outcrops and for the

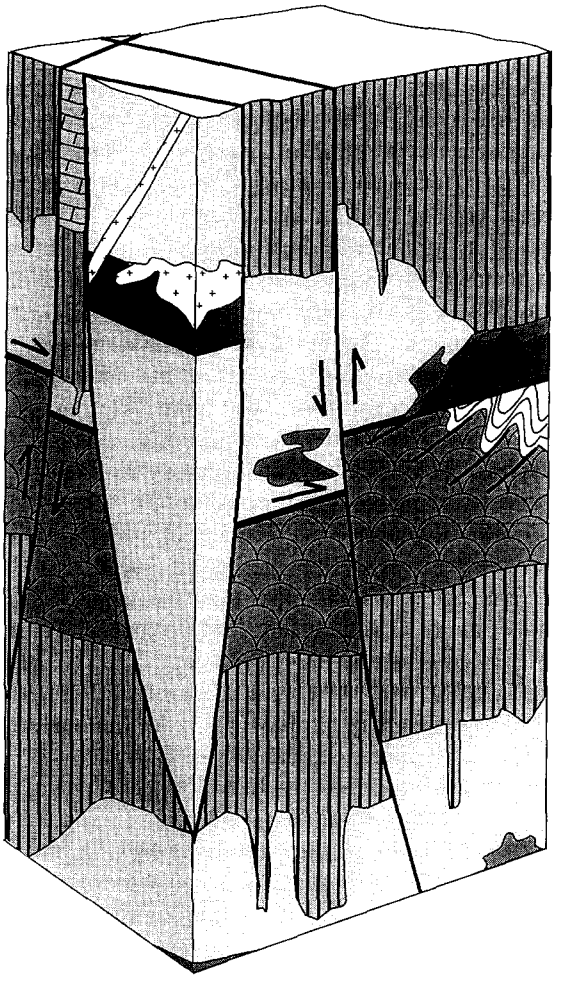


Fig. 2. Schematic column of the two ophiolite nappes, offset by normal faulting. Note the position of the extrusive rocks and associated sediments below the thrust. The two ophiolite nappes show the following particulars: upper nappe with granitoid intrusions and presence of mantle peridotites, the sheeted dikes commonly root in the peridotites and the thickness of the gabbro section is highly variable; lower nappe with a well-developed extrusive series.

uplift of the lower nappe in the southeastern sector of the island. These normal faults are not shown in Fig. 1.

3. Sampling and mineralogy

Eighty-four volcanic samples (Table 1) were taken in the well-exposed sheet flows and pillow and tube lavas from the top of the lower nappe with a water-cooled drill and oriented using both sun and magnetic compasses. The sample sites are located in Fig. 1 (sample 72 was lost at the outcrop). The sampled lavas are predominantly tholeiitic (Abbotts, 1981), aphyric, or contain plagioclase phenocrysts. They show a metamorphic overprint in prehnite-pumpellyite to greenschist facies with alteration of the matrix to chlorite (actinolite) in green lavas or to zeolite-smectite-hematite in red lavas (Abbotts, 1981). Magnetite-Ti-magnetite-maghemite and ilmenite are the dominant magnetic minerals in the rocks. Both minerals occur interstitially between plagioclase laths. Magnetite shows homogeneous colour under reflected light with local oxidation to maghemite-hematite. Ilmenite may show exsolution lamellae. Ilmenite or Ti-magnetite is commonly replaced by fine-grained rutile/anatase. Fine-grained hematite and iron-hydroxides are present in red lavas at the top of the pillow stack.

Sedimentary bedding of radiolarian cherts at the top of the extrusives was used to define the paleo-horizontal. The paleo-horizontal in the Shinzi area was defined by well-bedded tuffite layers. The tilt corrections listed in Table 1 are the average tilt of these overlying deep sea type sediments found closest to the sample localities. The variable tilt of the sediments along the island creates uncertainties for the corrections to apply to sample localities which are far from any sedimentary outcrop. Moreover, sediments in the Naft and Thumi area are commonly isoclinally folded with axial fold planes subparallel to the strongest dip of the sedimentary bedding and also subparallel to sheet flow and lava tube surfaces.

4. Methods and results

Standard size specimens were cut from the cores in the laboratory and the Normal Remanent Magnetization (NRM) of the lavas were measured using a CTF cryogenic magnetometer. All samples were stepwise thermally demagnetized in a μ-metal shielded furnace with 50° steps. After each heating step, the magnetic susceptibility at room temperature was monitored using a Bartington bridge. The components of magnetization were defined by principal component analysis (Kirschvink, 1980). Site means were calculated using Fisher (1953) statistics.

NRM intensities varied from 10^{-4} to 10^{-6} $A\,m^2\,kg^{-1}$. Submarine altered tholeiitic basalts have typical magnetic intensities in the order of 10^{-4} $A\,m^2\,kg^{-1}$ as is the case for most of the samples

Table 1
Site information and tilt corrections

Site		Tilt corr.		Sample	Lava type	Locality
(°N)	(°E)	Azimuth	Dip	(No.)		
20.18	58.65	000	00	1–2	A	South Peak
20.36	58.75	348	55	3–8	Th	Hakl
		029	50	9–13	Th	
20.41	58.80	253	59	14	Th	Centre
		288	66	15–16	Th	
		269	71	17–24	Th/A	
		292	82	59–85	Th	
20.51	58.89	178	60	25–31	Th	Thumi
20.57	58.91	125	37	32–35	A?	Shinzi
20.47	58.88	190	70	36–58	Th	Naft

A = alkaline; Th = tholeiitic.

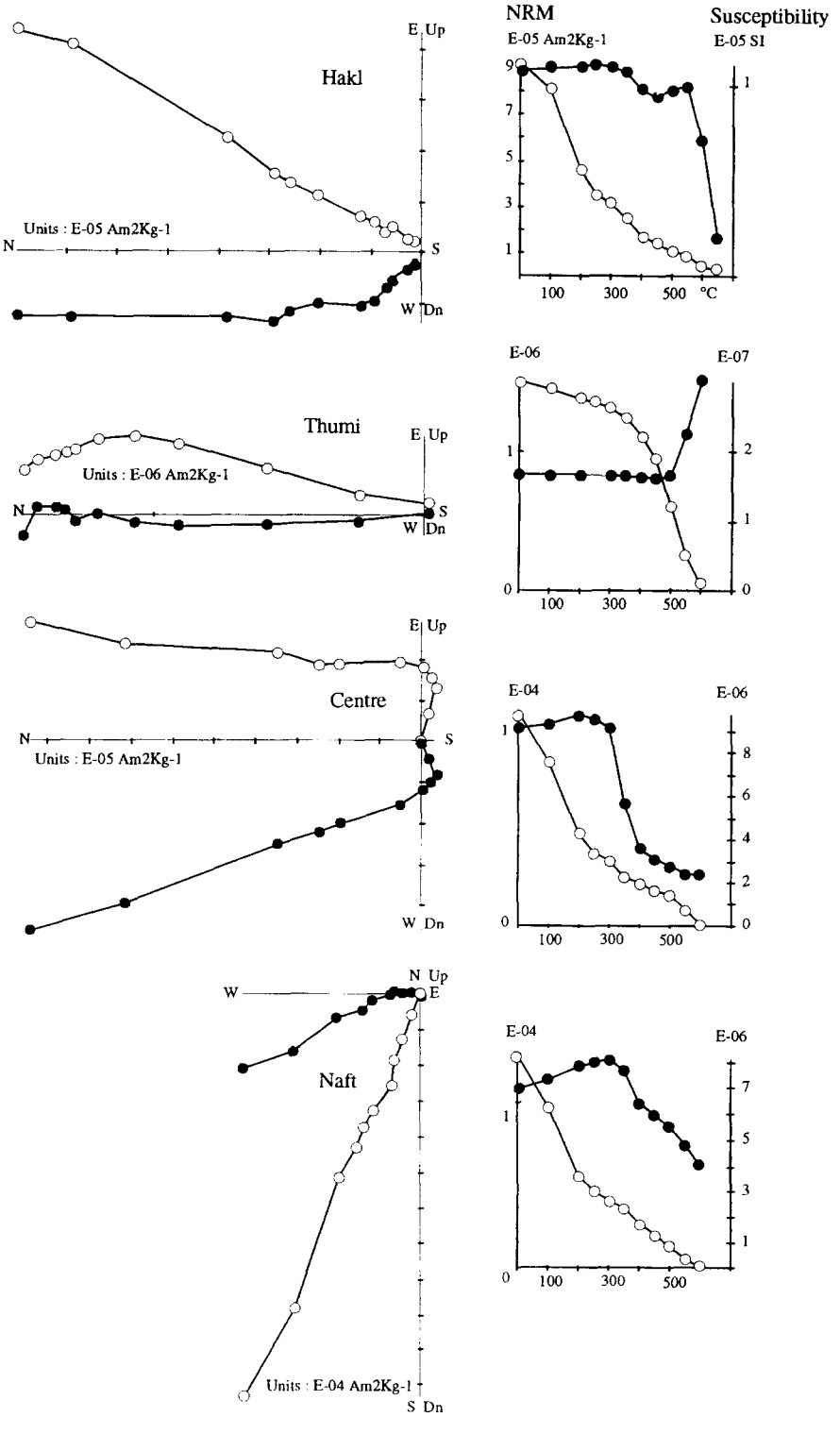

Hakl
E Up
Units : E-05 Am2Kg-1
N
S
W Dn
NRM
E-05 Am2Kg-1
Susceptibility
E-05 SI
9
7
5
3
1
1
100
300
500 °C
Thumi
E Up
Units : E-06 Am2Kg-1
N
S
W Dn
E-06
E-07
1
2
1
0
0
100
300
500
Centre
E Up
N
S
Units : E-05 Am2Kg-1
W Dn
E-04
E-06
1
8
6
4
2
0
0
100
300
500
N Up
W
E
Naft
Units : E-04 Am2Kg-1
S Dn
E-04
E-06
1
7
5
3
1
0
100
300
500

Table 2
Low-temperature mean directions of magnetization defined for Masirah

Site	N	In situ				Tilt corr.			
		I	D	k	α_{95}	I	D	k	α_{95}
Hakl	7	29.2	1.6	99	6.1	−22.0	3.4	53	8.4
Thumi	4	29.8	5.8	210	6.4	14.6	333.2	210	6.4
Centr	18	30.8	0.7	126	3.1	−2.2	340.1	33	6.1
Naft	21	31.3	358.7	101	3.2	68.0	234.7	101	3.2
Mean BT	50	30.7	0.4	113	1.9	33.6	327.9	3	16.1

N = number of samples; I = inclination; D = declination; k and α_{95} = precision parameter and 95% confidence cone of Fisher statistics. The mean directions of magnetization are given before (in situ) and after tilt correction.

analyzed here. Localities with magnetic intensities of 10^{-5} to 10^{-6} $Am^2 kg^{-1}$, as in the Thumi and Hakl areas, indicate additional chemical alteration of the magnetic mineralogy. Demagnetization curves (Fig. 3) have similar characteristics within one site but vary highly from site to site. A low-temperature component is commonly destroyed below 250–300°C at which point magnetization is reduced by half. This

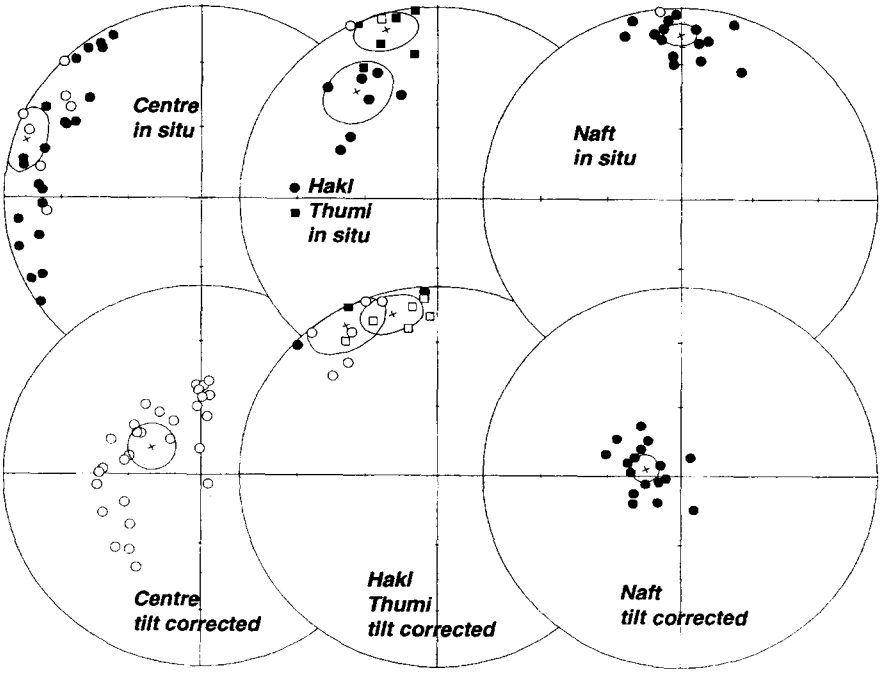


Fig. 4. Equal-area projections of mean high-temperature directions before and after tilt correction for Masirah sites. Filled and open symbols represent directions in the lower and upper hemispheres, respectively.

Fig. 3. Representative orthogonal diagrams (after tilt correction) and demagnetization curves for tholeiitic lavas. ● and ○ on the orthogonal diagrams represent, respectively, the horizontal and vertical planes. On the demagnetization curves, left axis (●) and right axis (○) represent, respectively, NRM intensity and room temperature susceptibility.

Table 3
High-temperature mean directions of magnetization defined for Masirah

Site	N	In situ				Tilt corr.			
		I	D	k	α_{95}	I	D	k	α_{95}
Hakl	8	31.9	324.0	17	13.7	−8.9	330.4	19	13.2
Thumi	7	9.8	343.2	35	10.3	−12.1	344.7	35	10.3
Hakl–Thumi	15	21.7	333.8	14	10.7	−10.5	337.1	21	8.5
Centre	29	7.4	288.7	7	10.5	−66.5	300.1	9	9.7
Naft	18	15.3	359.6	35	5.9	75.2	282.0	35	5.9

N = number of samples; I = inclination; D = declination; k and α_{95} = precision parameter and 95% confidence cone of Fisher statistics. The mean directions of magnetization are given before (in situ) and after tilt correction.

low-temperature remagnetization corresponds to either a viscous and/or a chemical magnetization acquired under the present-day magnetic field (Table 2). Samples from South Peak and Shinzi areas show only this stable present-day chemical remagnetization. For the other sites, after removal of this low-temperature component, a stable high-temperature direction can be defined above 450–500°C (Fig. 4).

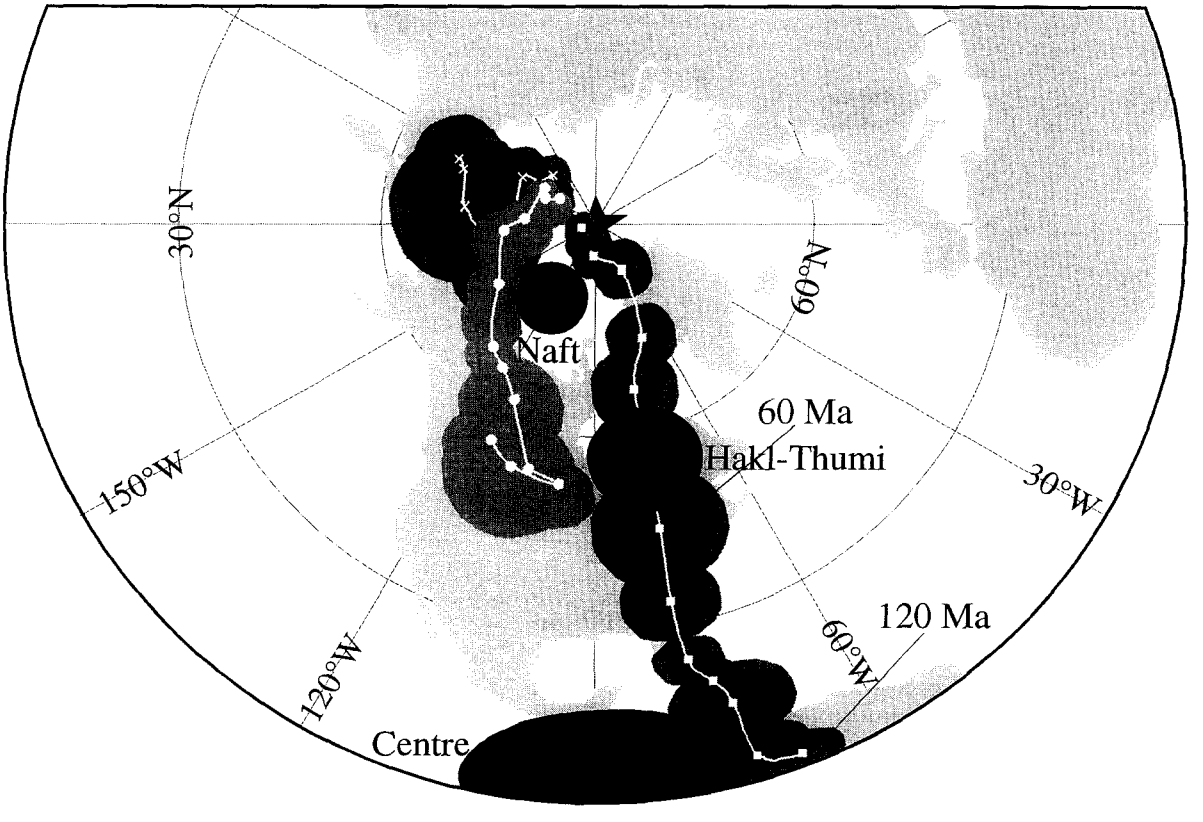


Fig. 5. Paleomagnetic poles (with their 95% confidence circles) determined from Masirah samples plotted together with the master polar wander paths proposed by Besse and Courtillot (1991) for the Indian (squares), Eurasian (crosses) and African (circles) continental plates. Equivalent azimuthal polar projection.

Table 4
Paleomagnetic poles and paleolatitudes defined for Masirah

Component		Lat.	Long.	k	α_{95}	Paleolat.
BT	IS	86	232	126	1.8	17 ± 3
HT Hakl–Thumi	TC	56	282	26	7.6	-5 ± 5
HT Centre	TC	3	271	4	14.6	-38 ± 12
HT Naft	IS	78	240	50	5.0	8 ± 5

The paleomagnetic poles (latitude, longitude) were calculated using in situ (IS) or tilt corrected (TC) mean directions of magnetization defined either at low (BT) or high (HT) temperature.

The mean high-temperature directions of magnetization are presented in Table 3 and Fig. 4. In some samples, an intermediate component seems to be present. This component, although it occurs typically between 350 and 500°C, shows widely dispersed directions, even within one site. This is interpreted as intermediate directions between the low- and high-temperature components.

Samples from the Hakl and Thumi areas show scattered but statistically similar high-temperature directions of magnetization (Table 3; Fig. 4), with both polarities in the Hakl samples. The two means for Hakl are: $I = -9.9°$, $D = 320.0°$, $k = 31$ ($\alpha_{95} = 16.8$, $N = 4$); and $I = 7.6°$, $D = 160.8°$, $k = 19$ ($\alpha_{95} = 21.4$, $N = 4$). The Hakl–Thumi mean direction is different from the high-temperature directions obtained for Centre (Table 3; Fig. 4). The Centre site includes three different sampling localities in which the magnetic directions overlap. Nevertheless, there is also a large scatter, mainly in declination. This scatter is likely to be due to local tilting with respect to the defined sedimentary paleohorizontal and may also explain why the fold test is not positive even though these two first components group better after tilt correction (see Table 3). A fairly well defined component, which is different from the previous

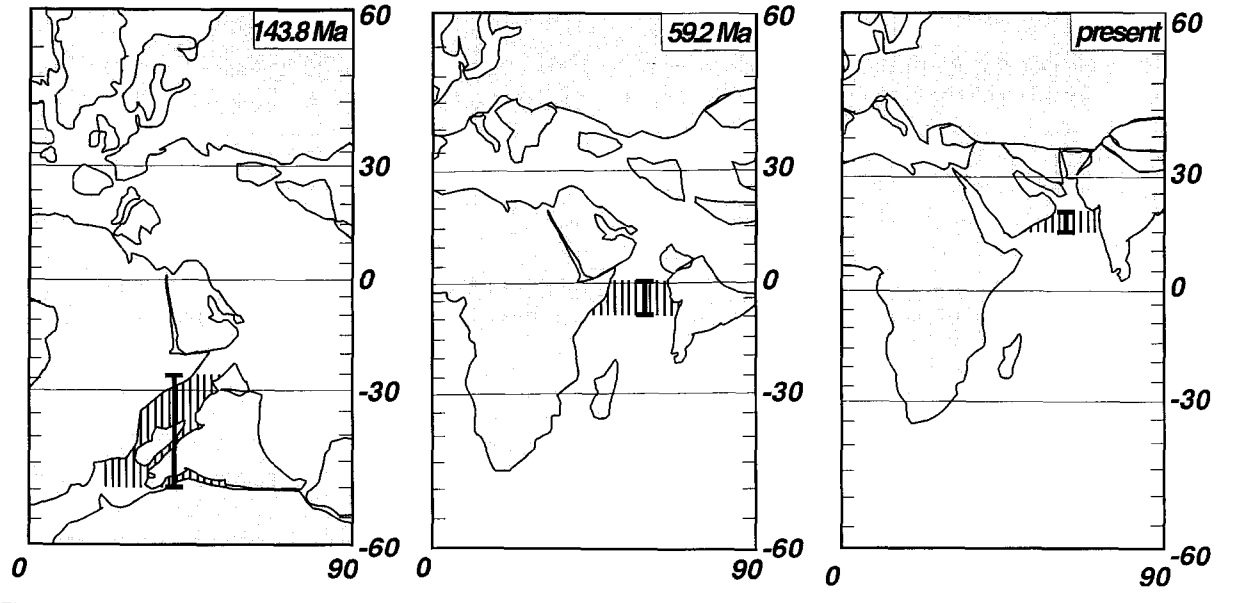


Fig. 6. Paleogeographic reconstruction showing the possible origin of the lower Masirah ophiolite nappe at latitudes of the West Somali basin, its latitude position during the emplacement of the second nappe, and present-day position. The paleogeographic reconstructions at 143.8 Ma, 59.2 Ma and Present are from Scotese et al. (1988).

ones, is found in the Naft area. Since only one tilt correction could be estimated in this area, there is no possible fold test. Nevertheless, it is impossible to correlate the paleomagnetic pole, which can be calculated from the tilt-corrected mean direction, with any polar wander path of interest in this area (e.g., Europe, Africa or India). This tends to indicate that this component must be acquired after the deformation phases or that a difference in tilting between pillows and sedimentary rocks occurred.

The paleopoles obtained from the Masirah samples are listed in Table 4 and presented in Fig. 5 together with the Master Polar Wander Paths (MPWP) for the Indian, African and Eurasian plates of Besse and Courtillot (1991). This figure shows that several sites have poles on or close to the Indian MPWP. If we assume that the determinations for these sites are correct, we can estimate the corresponding time of lava formation and remagnetization. The results for the lower Masirah nappe indicate an origin at a latitude of 38 ± 12°S, followed by a northward movement together with the Indian plate and a metamorphic event at a latitude around 5 ± 5°S sometime between 60 and 50 Ma (Fig. 6). This local reset of the paleomagnetic signal in lava of the lower nappe seems to be due to the southward directed emplacement of the upper Masirah ophiolite nappe (Marquer et al., 1995).

As long as we have no truly positive fold test, the possibility that all measured directions are secondary cannot be ruled out completely. This would mean that all volcanic rocks were remagnetized during folding and imbrication and subsequently affected by differential rotations (e.g., Parés et al., 1994), which could explain the dispersion in declination when the data are considered in situ. The striking coincidence of at least two of the tilt-corrected poles with the Indian MPWP makes us prefer the first interpretation.

5. Conclusions

The two-nappe structure of Masirah island and the paleomagnetic indications reported in this study suggest that Masirah is neither a laterally displaced part of the Semail ophiolite nor an in-situ uplift of the Indian Ocean floor. The high-temperature directions of several sites indicate an origin at latitudes of 38 ± 12°S. The evolution, however, between its formation in the Tithonian (around 145 Ma) and the emplacement of the second nappe around the Cretaceous/Tertiary boundary seems to be related to the Indian plate. A change in the direction of movement of the Indian plate around 125 Ma as a result of the initial opening of the Bengali oceanic basin between Antarctica and India was proposed by Owen (1983). This opening initiated rifting between the Indian plate and the Madagascar microplate which is probably represented on Masirah by a tectonic phase associated with alkaline magmatic activity at 123 ± 2 Ma (Nägler and Frei, 1994; Immenhauser, in prep.). Comparing different paleogeographic reconstructions for the time period 150–120 Ma there is a variation in paleo-longitude as large as 60° (Savostin et al., 1986; Scotese et al., 1988; Powell et al., 1988) for the position of India. This uncertainity results in a highly variable size of the oceanic basin between Africa and Madagascar and Madagascar and India. The results confirm that between the late Maastrichtian and the pre-Eocene a compressive event caused the obduction of the upper onto the lower ophiolite nappe. This is probably slightly prior to the final obduction of the lower ophiolite nappe onto the Arabian continental margin. The youngest Fayah sediments of the Batain area (Shackleton et al., 1990) are of Maastrichtian age similar to Masirah island. However, it is not clear if the Fayah formation is a special facies of the Aruma Group and related to the area of the Oman Mountains and the Semail Ophiolite (e.g., Shackleton et al., 1990; Béchennec et al., 1992) or if it is of allochthonous origin, related to the "Eastern Ophiolite belt". Supporting indications for a compressive event between the Indo-Pakistani and the Afro-Arabian plate at that time are found in Pakistan where similar obduction ages were obtained for the Muslim Bagh and the Bela ophiolites (Allemann, 1979; Mahmood et al., 1995). This geological evidence may indicate that the 95% confidence bar from the paleomagnetic measurements shown in Fig. 6 (59.2 Ma) could be underestimated or that the Fayah Formation was deposited south of its present position (e.g., in the area of 5 ± 5°S). These conclusions may further indicate that the western outline of the Indo-Pakistani plate was different in shape to the one generally used

in paleogeographic reconstructions, or that the Indo-Pakistani plate moved closer to the Afro-Arabian plate in the Late Cretaceous. The obduction of the Semail ophiolite onto the Arabian continental margin seems to have occurred earlier (e.g., Allemann and Peters, 1972; Glennie et al., 1974; Tilton et al., 1981; Tippit et al., 1981). This would indicate that where the ophiolite belts of the Oman Mountains (Semail ophiolite) and the "Eastern Ophiolite belt" overlap (e.g., Batain Coast), the Semail ophiolite would represent the lower unit. However, the downflexure of the Arabian continental crust as a result of the obduction of the Semail ophiolite could have facilitated the obduction of the "Eastern Ophiolite belt".

Acknowledgements

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